RETRO ARCADE GAME REVIVAL - SNAKE

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Outline

For our game, we have decided to recreate the classic arcade game ‘Snake’. Snake involves a player moving a character around a map using the arrow keys, eating apples to gain points and grow longer while avoiding running into walls or its own body. We are hoping to recreate this game and its scoring system, while adding to its aesthetic value by adding background music, sound effects for eating and dying and having unique and creative retro-style apple, snake and background sprites. We will also be adding a challenge by putting in extra walls to dodge while playing. Additionally, we will also be giving the game a new style, by replacing the snake with a worm and changing the scenery. The main goal of the project is to recreate the original snake’s functions and gameplay loop, while also giving it a more colourful style and new features that will improve the game’s ability to allow the user to play again, increase difficulty and track and save high scores to a database.

Design and Implementation

Our snake game features many design choices that enhance user experience while playing. We have implemented a pixelated dirt background to tie into the retro theme found throughout many retro arcade games, making the game feel like a genuine game from the 80’s. Our ‘Snake’ is a block design, similar to the one used in the original game. This not only allows for a more retro feel, but also makes it clearer as to where to turn in order to collect an apple. We have also incorporated sound effects into our game for when the user starts the game, eats an apple and loses, as well as background music, all of which fit the retro style that we are trying to achieve. This feature both ties into the retro theme of the game, but also makes us a unique version of the classic game.

Our code uses simple methods for the movement and mechanics of the snake, continuously moving one square every tick, with tick rate increasing as difficulty went up, making the snake faster, and displaying it on the screen for the user to see and react to by turning it with the arrow keys. When the snake eats an apple, it grows longer. We achieved this by creating a system where when the snake head is on the same game cell as an apple, a new head is generated/drawn, to go with the rest of the snake. The score then increases by 1 to create the scoring system and after the player loses, the code checks if the new score is greater than the high score and if it is, it saves it to a database.

Challenges and Solutions

Challenge 1
A major challenge that was faced during the project was creating an effective ‘play again’ system, that would allow the user to click a button after they died and let them start again without kicking them out of the game. We struggled to achieve this due to the code not acknowledging the difference between the ‘Play again’ and ‘Quit’ buttons.

Solution
This problem was solved by modifying the game loop to check if a certain criteria was met, in this case, was the ‘Quit’ button clicked, and if it wasn’t, restart the game with the ‘Play Again’ button.

Challenge 2
Another major challenge that was faced during the project was creating a scoring system that would accurately tally the users score as they played, adding a point every time the user eats an apple and displaying the updated score immediately. When first trying to achieve this, the game would crash after the user ate an apple because the game wanted to reset and refresh itself every time it would update the score.

Solution
To solve this problem, we first brainstormed ideas on how to improve this function and performed a lot of trial and error. We finally figured out that by copying and modifying our already existing code used in displaying the countdown in the start sequence, we could display the scoreboard by adding a variable that changed the score when the user eats an apple. Additionally we moved the positioning of the code so it could be called at the correct place in the code, reducing crashes.

Challenge 3

A third major challenge that was faced during the project was coming up with ideas to improve and add onto our game. This was a problem because we got to a point where our game was technically complete, but had no unique points and wasn't fun to play.

Solution

This problem was solved by researching online and gaining inspiration from similar projects and games that already existed, finding which features made the game fun. We added a system that tracked high scores as well as multiple difficulty levels. This allows players to enjoy the game more and provides more to do, as they can choose to play at their skill level and a sense of challenge and accomplishment is added as players try to beat the highscore.

Accessibility and Inclusivity

Accessibility and inclusivity are addressed in the game through a variety of thoughtful features designed to ensure that everyone can play, while also enhancing the overall quality of the gameplay experience.

- The game uses high-contrast colors to make key elements like the snake, food, and background clearly distinguishable through the use of bright green, dark brown and light pink. This not only improves general visibility for all players but also supports those with visual impairments or colour blindness.
- Essential information, such as the game-start countdown, is communicated through both audio and visual cues. For example, a sound plays alongside a prominently displayed on-screen countdown. This dual-channel feedback ensures that players who may be deaf or hard of hearing, or those in noisy environments, are not disadvantaged.
- Different difficulty levels were made so that players of all skill levels can enjoy the game and challenge themselves.
- There is limited text, allowing for non-English speakers to play easily.

Ethical Considerations

We have included many components in our game that makes it a fun and inclusive environment for everyone. Our game has no violence of any kind and no graphics that may be triggering such as blood or death. Additionally, our game does not contain any plagiarised work as all imported material (sounds, Sprites) is open for free use or was designed by our group and we have added enough unique points, such as difficulty modes to our game so that it is not an exact copy of the original snake game. We have also implemented no features that would exclude or make someone feel left out, ensuring a fun game, suitable for everyone to play.

Conclusion

The project successfully managed to make a unique adaptation of the arcade game “Snake” with a new style and aesthetic themed around a worm. The addition of new mechanics such as a high-score and multiple difficulty modes encourages replayability and user engagement. Additionally the game is accessible by many users as it has high contrast visuals , audio cues and different difficulty levels, ensuring players of all abilities and skill levels can enjoy the game. Areas of the project that could be improved include adding additional features, such as the scrapped wall feature and powerups to add more unique flair and engagement to the game, making sprites more detailed and responsive (adding a head to the snake that turns with player movement). Adding these additional features to the game would make it more fleshed out and unique in its style.

References

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